

9	(a)	(i)	lowest frequency of electromagnetic radiation	M1
			giving rise to emission of electrons (from the surface)	A1
		(ii)	threshold frequency = $(9.0 \times 10^{-19}) / (6.63 \times 10^{-34})$	M1
			= 1.4×10^{15} Hz	A1
	(iii)	<i>either</i>		
			frequency of radiation between 5×10^{14} Hz and 10×10^{14} Hz	
			or energy of photons between 3.3×10^{-19} J and 6.6×10^{-19} J	
			or threshold wavelength: zinc = 340 nm, sodium = 520 nm, platinum = 220 nm	M1
			emission from sodium and zinc	A1
	(iv)	1.	photon interact with electron below surface	M1
		energy required to bring electron to surface	A1	
2.		show threshold frequency is 5.8×10^{14} Hz		
		e.g extrapolate graph to intersect with x-axis		
		e.g chose a point (7.0, 0.5) and substitute into $E_{\text{MAX}} = \frac{h}{e}(f - f_0)$	M1	
		metal is sodium	A1	
(b)	(i)	1.	centripetal force of orbiting electron provided by electric force acted by proton	
			and so $\frac{mv^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}$	M1
			correct manipulation to obtain $\frac{1}{2}mv^2 =$ expression	A1
		2.	$E_T = \frac{e^2}{8\pi\epsilon_0 r} + \left(-\frac{e^2}{4\pi\epsilon_0 r}\right)$	M1
			$E_T = -\frac{e^2}{8\pi\epsilon_0 r}$	A0
(ii)		$\lambda = \frac{h}{\sqrt{2mE_K}} = \frac{h}{\sqrt{2m \times \frac{e^2}{8\pi\epsilon_0 r}}}$	M1	
		$\lambda = \frac{h}{e} \sqrt{\frac{4\pi\epsilon_0 r}{m}}$	A0	
(iii)		$E_T = -\frac{e^2}{8\pi\epsilon_0 \times \frac{n^2 h^2 \epsilon_0}{\pi m e^2}} = -\frac{m e^4}{8 \epsilon_0^2 h^2} \frac{1}{n^2}$	B1	

$$k = \frac{me^4}{8\epsilon_0^2 h^2} = \frac{(9.11 \times 10^{-31})(1.60 \times 10^{-19})^4}{8(8.85 \times 10^{-12})^2 (6.63 \times 10^{-34})^2} \quad \text{M1}$$

(iv) $k = 2.17 \times 10^{-18} \text{ J}$ A1

$$\lambda = \frac{hc}{\Delta E}$$

longest wavelength for transition from $n = 3$ to $n = 2$ and shortest wavelength from $n = \infty$ to $n = 2$ C1

correctly show longest wavelength is 660 nm M1

correct show shortest wavelength is 367 nm M1

so, Balmer series are visible A0